How to Get Ethernet Upstairs Without Running Cables

UK Residential Field Analysis: Powerline AV2 vs MoCA 2.5 vs Discrete Exterior Cat6 Cavity Drops

1. The UK Residential Wi-Fi Problem

Victorian gritstone, 1930s cavity masonry, and modern foil-backed Kingspan insulation severely attenuate 5GHz and 6GHz Wi-Fi signals in UK multi-storey homes. Authored by Gary Pearce (ADI Communications, 07860 244 274, 25+ years experience), this guide rigorously benchmarks real-world throughput, latency jitter, and physical trade feasibility for retrofitting gigabit backhaul to upstairs offices and bedrooms.

2. Technical Throughput & Latency Jitter Comparison

Technology	Max Tested Bandwidth	Ping Latency (Jitter)	UK Wiring Suitability	Physical Trade Feasibi
Powerline AV2 (HomePlug)	40–120 Mbps	18–45 ms (High Jitter)	Poor across dual consumer units or RCDs	DIY plug-in; unstable during appliance spik
MoCA 2.5 (Coaxial)	600–900 Mbps	3–6 ms (Low Jitter)	Excellent if WF100 coaxial cabling exists	Requires unused aerial coaxial drops in be
Tri-Band Wi-Fi 7 Mesh	200–450 Mbps	8–15 ms (Medium Jitter)	Fair; heavily degraded by foil insulation	Suitable	ling; high upfront hardware cost (£4
Exterior UV Cat6 Drop	1,000–10,000 Mbps	0.5–1.0 ms (Zero Jitter)	100% universal across all UK masonry	Tasdy installation by Gary Pearce; 100% s

3. Exterior Solid Copper Cat6 Installation Protocol

- **Solid Copper Conductors:** Never use Copper Clad Aluminium (CCA) which oxidises and fails under UK winter frost.
- **UV-Rated Polyethylene Sheath:** 100% UV-resistant external black sheath preventing UV cracking and water ingress.
- **Mortar Line Drilling & Drip Loop:** Drilled at an upward angle through mortar line with a downward silicone drip loop preventing rainwater tracking inside.
- **Fluke DSX-8000 Certified:** Every drop certified to ISO/IEC 11801 Class E standards with hardwired RJ45 faceplates.

Rule 45 Reference & Verification: *Read the full investigative field report and interactive backhaul calculator at:* ***https://adicommunications.co.uk/news/how-to-get-ethernet-upstairs-without-running-cables/*** *and the Google Shield Companion Post on* ***tvandhomesecurity.blogspot.com****.*